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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2022
Department of Computer Science and Engineering
CSE 4221
Natural Language Processing

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the immediate right column of the questions indicate full marks.
iii) The rightmost column indicates course outcomes.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Define corpus. What are the properties of a corpus? (05) [CO1]
   b) "Patterns are greedy" – why? Design a regular expression to answer the following statement: "Find a computer with at least 6 GHz and 512 GB RAM for less than BDT 4000". (10) [CO1]
   c) Define subword tokenization. Explain the BPE algorithm for subword tokenization. Use the following corpus to explain the algorithm.
   "low low low low low lowest lowest Newer Newer Newer Newer wider wider wider new new"
   Test Set: newer, lower. (15) [CO2]
   d) Define stop words. What are the applications of stop word detection in NLP tasks? (05) [CO1]

2. a) You have two strings "brief" and "drivers". Figure out whether "drive" is closer to "brief" or "drivers" using minimum edit distance algorithm. (14) [CO2]
   b) What is *n*-gram language model? Find the uni-gram, bi-gram, and tri-gram from the following corpus.
   "This is Big Data AI Book". (06) [CO3]
   c) Why do you need smoothing techniques for *N* gram model? Explain the following smoothing techniques: (15) [CO3]
      i) Add-1 smoothing.
      ii) Stupid back off.
      iii) Kenser-Ney smoothing.

3. a) Give a formal definition of supervised text classification. How can you differentiate generative and discriminative classifier? (05) [CO4]
   b) What is "Bag of words" model? What are the assumptions of Bag of words model? How does Bag of words applied for multinomial Naïve Bayes Classifier? Using the following training set find the class of the test set. (15) [CO4]

		Document	Class
Training	:	Just plain boring	–ve
		No surprise and few laughs	–ve
		Very powerful	+ve
		Most fun film of the summer	+ve
Test	:	Predictable with no fun	?

   c) When is "Accuracy" not a good metric? Explain with example. (07) [CO3]
   d) "The noisy channel is a kind of Bayesian inference" – explain the term. (08) [CO3]

4. a) What are the types of spelling errors? you wanted to type 'three' but mistakenly you typed 'there'. How can you detect such an error automatically? How to correct these types of errors? (12) [CO3]
   b) What are the common structural ambiguities of Natural Language Processing? Explain. (06) [CO3]
   c) What is partial parsing? What are the applications of partial parsing? (05) [CO3]
   d) Consider the following grammar G. Using the CYK algorithm, find whether "ababa" is L(G) or not. (12) [CO3]

$$S \rightarrow AB|BC$$
$$A \rightarrow BA|a$$
$$B \rightarrow CC|b$$
$$C \rightarrow AB|a$$

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Explain the large language model (LLM) with an example. Discuss computational resources and energy consumption in LLM. (10) [CO1]
   b) "Parts-of-Speech (POS) tagging is a disambiguation task" – justify the statement with example. (10) [CO2]
   c) Discuss POS tagging and tagsets for English. (10) [CO3]
   d) How difficult is POS tagging in English? (05) [CO3]

6. a) Discuss Conditional Random Fields using Bayes' rule. (10) [CO2]
   b) We have a dataset (NewG). Some information about the dataset is given below. (25) [CO2]

No. of VB	:	17619	Starting probability VB	:	0.024
No. of MD	:	11338	Starting probability MD	:	0.11
No. of NN	:	16559	Starting probability NN	:	0.86
$t_{i-1} - t_i$			$t_i - w_i$		
VB – VB	:	8583	VB – race	:	5391
VB – MD	:	6988	VB – will	:	6308
VB – NN	:	5826	VB – begin	:	7488
MD – VB	:	7984	MD – race	:	6557
MD – MD	:	4204	MD – will	:	6635
MD – NN	:	4046	MD – begin	:	6905
NN – MD	:	5006	NN – race	:	4163
NN – NN	:	4568	NN – will	:	6103
			NN – begin	:	7863

Use the dataset to answer the following questions:

i) Calculate the transition probability (A).
ii) Calculate the emission probability (B) for sentence: "race will begin".
iii) Find the tag sequence for the sentence "race will begin" using hidden Markov model.

7. a) Discuss the sequence to sequence model with attention mechanism (AM). Demonstrate the limitations without AM. (10) [CO2]
   b) Can hidden Markov models incorporate features? (05) [CO2]
   c) What is bootstrapping in information extraction? Explain the architecture for bootstrapping. (10) [CO4]
   d) How do we detect events? Write down features commonly used in both rule-based and statistical approaches to event detection. (10) [CO4]

8. a) Explain term weighting and document scoring. (05) [CO5]
   b) (12) [CO5]

Query	:	Hello dear
doc1	:	Hello my friends
doc2	:	Hello Karim, hello Rahim
doc3	:	My dear friend, Hello
doc4	:	Hello, Dear, dear

Using term weighting and document scoring, rank the document for Query.

   c) Discuss language models (LMs) to do question answering (QA). Write down the comparative analysis between LM QA models and classic QA models. (10) [CO5]
   d) What is homograph disambiguation? What problems does CMU face in this regard, and how does UNISYN address these issues? (08) [CO4]